

Chapter 2

Wave Functions and Observables

Lecture 4: Wave functions continued

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Lecture 3 introduced the wave function $\psi(x)$ as the state of a particle in one dimension. We now give the wave function its probabilistic meaning and study how superposition and Fourier analysis connect position and momentum.

This remains a first version of the quantum rules. In three dimensions a one-particle state is $\psi(x, y, z)$, and a two-particle state depends on both particle positions. For now, we continue with one particle moving in one dimension so that the essential ideas remain visible.

Questions.

- How does $\psi(x)$ determine the probabilities of position measurements?
- Why must probability amplitudes be added before probabilities are calculated?
- How can a general wave function be built from definite-momentum plane waves?
- Why does better localization in position require a wider range of momenta?

2.2 The probability interpretation

2.2.1 The Born rule

Postulate (Probability density). If a particle is in the state $\psi(x)$, then the probability density for finding it at position x is

$$\rho(x) = |\psi(x)|^2 = \psi^*(x)\psi(x). \quad (2.9)$$

The absolute square is essential. The wave function may be complex, so it cannot itself be a probability. The quantity $|\psi(x)|^2$, however, is real and nonnegative.

Position is a continuous variable. The probability of finding the particle at

one exact point is generally zero; meaningful probabilities refer to intervals. For a very small interval,

$$P(x \leq X \leq x + dx) = |\psi(x)|^2 dx, \quad (2.10)$$

while for a finite interval $[a, b]$,

$$P(a \leq X \leq b) = \int_a^b |\psi(x)|^2 dx. \quad (2.11)$$

A position measurement on one particle produces one definite result. The density $|\psi|^2$ predicts the distribution obtained from many identically prepared particles.

2.2.2 Normalization

The particle must be found somewhere in the allowed region. Therefore a normalized wave function satisfies

$$\int_{\text{all } x} |\psi(x)|^2 dx = 1. \quad (2.12)$$

For a particle on the real line, “all x ” means $(-\infty, \infty)$. For a confined particle, the integral extends only over the allowed region.

If a given function $\psi(x)$ is not normalized but has a finite norm, define

$$N = \int_{\text{all } x} |\psi(x)|^2 dx. \quad (2.13)$$

Then

$$\psi_N(x) = \frac{\psi(x)}{\sqrt{N}}, \quad \rho(x) = |\psi_N(x)|^2 = \frac{|\psi(x)|^2}{N}. \quad (2.14)$$

Remark (Dimensions). Probability is dimensionless, so $|\psi(x)|^2 dx$ must be dimensionless. In one dimension,

$$[\psi] = L^{-1/2}. \quad (2.15)$$

The dimensions change with the number of spatial coordinates: for example, a one-particle wave function in three dimensions has dimension $L^{-3/2}$.

2.2.3 Reasonable wave functions

A physical one-particle wave function should be single-valued and normalizable. For the ordinary finite potentials considered first, it should also be continuous and sufficiently smooth for the derivatives appearing later in the theory to exist. These conditions have physical content:

- A multi-valued curve does not assign a unique amplitude to each position.
- A function with $\int |\psi|^2 dx = \infty$ cannot be normalized as an ordinary state.

- A discontinuity or singular derivative can make quantities such as momentum or energy ill-defined.

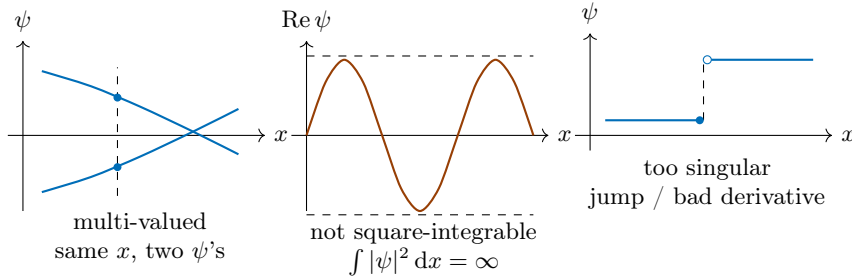


Figure 2.3: Three curves that are not ordinary physical wave functions for the finite-potential problems studied first.

Exact plane waves on the infinite line are also not normalizable. Nevertheless, they are useful idealized states, just as perfectly definite positions will later be represented by idealized delta functions. Normalizable wave packets describe physically preparable states.

2.3 Superposition and interference

2.3.1 The superposition principle

Postulate (Superposition principle). If $\psi_1(x)$ and $\psi_2(x)$ are possible states, then a normalized linear combination

$$\psi(x) = \alpha\psi_1(x) + \beta\psi_2(x), \quad \alpha, \beta \in \mathbb{C}, \quad (2.16)$$

is also a possible state.

A superposition is a new quantum state, not ordinary ignorance about which state the particle “really” occupies. The distinction becomes visible because amplitudes can interfere.

For $\psi = \alpha\psi_1 + \beta\psi_2$,

$$\begin{aligned} |\psi|^2 &= (\alpha\psi_1 + \beta\psi_2)^*(\alpha\psi_1 + \beta\psi_2) \\ &= |\alpha|^2|\psi_1|^2 + |\beta|^2|\psi_2|^2 + \alpha^*\beta\psi_1^*\psi_2 + \alpha\beta^*\psi_1\psi_2^*. \end{aligned} \quad (2.17)$$

The last two terms are interference terms. Their sum is real, but it may be positive or negative, producing enhanced probability in some regions and reduced probability in others.

Proposition 2 (Amplitudes first). Quantum mechanics adds complex probability amplitudes before calculating probabilities:

$$\boxed{\psi = \psi_1 + \psi_2 \implies P \propto |\psi_1 + \psi_2|^2.} \quad (2.18)$$

In general, this is not equal to $|\psi_1|^2 + |\psi_2|^2$.

This is the mathematical origin of interference in the double-slit experiment and in experiments where paths are recombined. If information is introduced that distinguishes the alternatives, the interference pattern can change or disappear.

2.3.2 Localized states and plane waves

A state sharply peaked near x_0 gives a narrow position distribution and hence good position information. A plane wave,

$$\psi_k(x) = Ae^{ikx}, \quad (2.19)$$

has

$$|\psi_k(x)|^2 = |A|^2, \quad (2.20)$$

which is independent of position. It has a definite wave number and momentum,

$$\lambda = \frac{2\pi}{|k|}, \quad p = \hbar k, \quad (2.21)$$

but no position localization. The sign of k gives the direction of the momentum, while $|k|$ determines the wavelength.

Now consider two localized packets:

$$\psi(x) = \alpha\psi_{x_1}(x) + \beta\psi_{x_2}(x). \quad (2.22)$$

The particle may be detected in either region. If the packets barely overlap, $|\alpha|^2$ and $|\beta|^2$ give the two probabilities after normalization. The average position can lie between x_1 and x_2 , even though no individual detection occurs at the average.

For two plane waves,

$$\psi(x) = e^{ik_1x} + e^{ik_2x}, \quad (2.23)$$

the probability density is

$$\begin{aligned} |\psi(x)|^2 &= (e^{-ik_1x} + e^{-ik_2x})(e^{ik_1x} + e^{ik_2x}) \\ &= 2 + 2\cos[(k_2 - k_1)x]. \end{aligned} \quad (2.24)$$

Although each plane wave separately has uniform probability density, their superposition has spatial structure.

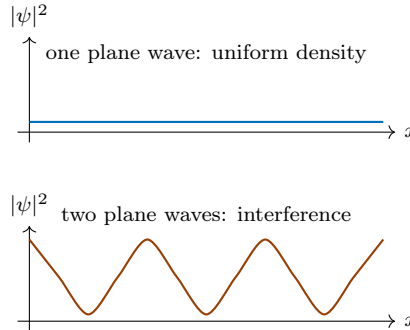


Figure 2.4: Superposition can turn two uniform plane-wave densities into a position-dependent interference pattern.

2.3.3 Many plane waves make a wave packet

Adding more plane waves gives more control over the interference pattern. With a suitable range of wave numbers and phases, many delocalized plane waves can combine constructively near one position and destructively elsewhere, producing a localized wave packet.

The price of position localization is a spread in momentum. One plane wave has one k and hence one momentum $p = \hbar k$. A localized packet requires many values of k , so it contains many possible momenta:

$$\begin{aligned} \text{one precise } k \text{ and } p &\implies \text{spread out in } x, \\ \text{localized in } x &\implies \text{many } k\text{'s and uncertain } p. \end{aligned} \quad (2.25)$$

This is the Fourier origin of the position-momentum uncertainty relation. The precise definitions of Δx and Δp will be introduced in the next stage of the course.

2.4 Fourier analysis and the momentum basis

2.4.1 Fourier transform

Fourier analysis makes the preceding picture precise. A sufficiently well-behaved function on the real line can be expressed as a continuous superposition of plane waves:

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk. \quad (2.26)$$

The coefficient function is

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx. \quad (2.27)$$

The two functions $f(x)$ and $\tilde{f}(k)$ contain the same information. Each can be reconstructed from the other.

Remark (Series versus transform). A Fourier series uses discrete wave numbers and is natural for periodic functions or a finite periodic box. A Fourier transform uses a continuous range of k and is natural on the whole real line.

For a wave function,

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\psi}(k) e^{ikx} dk. \quad (2.28)$$

Because e^{ikx} describes definite momentum $p = \hbar k$, the plane waves form the momentum basis. The coefficient $\tilde{\psi}(k)$ is the complex probability amplitude of the momentum component with wave number k .

2.4.2 Momentum-space probability

With the Fourier convention above, Parseval's theorem gives

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} |\tilde{\psi}(k)|^2 dk. \quad (2.29)$$

For a normalized state, both integrals equal one. Therefore

$$P(k \leq K \leq k + dk) = |\tilde{\psi}(k)|^2 dk. \quad (2.30)$$

Momentum is $p = \hbar k$, so $dk = dp/\hbar$. If we continue to use $\tilde{\psi}(k)$, then

$$P(p \leq P \leq p + dp) = \left| \tilde{\psi}\left(\frac{p}{\hbar}\right) \right|^2 \frac{dp}{\hbar}. \quad (2.31)$$

The factor $1/\hbar$ is required because probability density changes when the variable changes from k to p .

Proposition 3 (Fourier origin of uncertainty). A narrow distribution in position requires a broad range of Fourier wave numbers. Since $p = \hbar k$, a state cannot be arbitrarily narrow in both position and momentum:

$$\Delta x \Delta p \gtrsim \hbar. \quad (2.32)$$

Later we will derive the exact inequality $\Delta x \Delta p \geq \hbar/2$.

2.5 Randomness and measurement statistics

The Geiger-counter demonstration gives a concrete example of reproducible randomness. Near ordinary surroundings, a Geiger counter occasionally clicks because of background radiation. Near uranium glass, its average click rate increases because the glass contains uranium. Individual decay times remain unpredictable even though the average rate is reproducible.

Quantum mechanics predicts a probability distribution or average rate, not the exact outcome of every individual event. This is the same statistical structure expressed by the Born rule: the wave function gives reproducible probabilities for an ensemble of identically prepared systems, while a single measurement produces one definite but generally unpredictable result.

Questions.

1. Why is $|\psi(x)|^2$ a probability density rather than a probability?
2. Why is an exact plane wave useful even though it cannot be normalized on the whole real line?
3. If $k_2 > k_1 > 0$, which plane wave has the shorter wavelength and larger momentum?
4. What happens to the range of momenta when a wave packet is made narrower in position?

5. Why is $|\psi_1 + \psi_2|^2$ generally different from $|\psi_1|^2 + |\psi_2|^2$?

Lecture 4 summary

- The Born rule assigns the position probability density $\rho(x) = |\psi(x)|^2$.
- A physical state is normalized so that the total probability of finding the particle is one.
- Quantum states obey superposition, and cross terms in $|\psi|^2$ produce interference.
- Plane waves have definite k and momentum $p = \hbar k$, but they are completely delocalized in position.
- The Fourier transform expresses a wave function as a superposition of momentum components. The density $|\tilde{\psi}(k)|^2$ predicts momentum measurements.
- Position localization requires many wave numbers and therefore a spread of momenta. This is the Fourier origin of uncertainty.
- Quantum randomness is reproducible statistically even when individual outcomes are unpredictable.

Suggested reading.

- Feynman, Leighton, and Sands, *The Feynman Lectures on Physics*, Vol. III, Ch. 2, *The Relation of Wave and Particle Viewpoints*.
- Review Lecture 3 on the de Broglie relations and the wave function as the quantum state.